



Факультет Программной Инженерии и Компьютерной Техники

Лабораторная работа №2
«Создание взаимосвязанной IP-сети»
по дисциплине
«Администрирование систем и сетей»
Желаем получить по «Е» баллу

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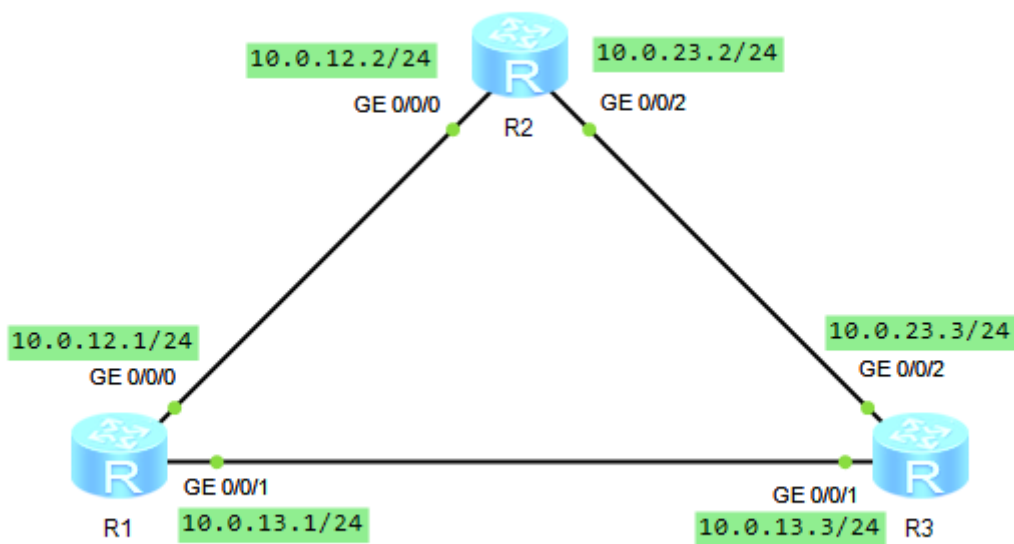
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Цели.

Лабораторная работа помогает получить практические навыки по изучению следующих тем:

- Процедура настройки IPv4-адреса на интерфейсе
- Функции и значение loopback-интерфейсов
- Принципы генерирования прямых маршрутов
- Процедура настройки статических маршрутов и условия, при которых используются статические маршруты
- Процедура проверки возможности установления соединения сетевого уровня с помощью инструмента ping
- Процедура настройки статических маршрутов и сценарии их применения

Топология сети



Процедура конфигурирования

Шаг 1 Настройте основные параметры устройств.

Задайте имена устройствам.

R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sys
[Huawei]sysname R1
[R1]
```

R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sys
[Huawei]sysname R2
[R2]
```

R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sys
[Huawei]sysname R3
[R3]
```

Шаг 2 Выведите на экран IP-адрес текущего интерфейса и таблицу маршрутизации маршрутизатора.

Выведите на экран статус интерфейса на маршрутизаторе (в данном случае на примере R1).

R1

```
[R1]display ip i
[R1]display ip interface b
[R1]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 1
The number of interface that is UP in Protocol is 1
The number of interface that is DOWN in Protocol is 3

Interface                IP Address/Mask    Physical  Protocol
GigabitEthernet0/0/0      unassigned         up        down
GigabitEthernet0/0/1      unassigned         up        down
GigabitEthernet0/0/2      unassigned         down      down
```

NULL0	unassigned	up	up(s)
-------	------------	----	-------

Выведите на экран таблицу маршрутизации на маршрутизаторе (в данном случае на примере R1)

```
[R1]disp
[R1]display ip
[R1]display ip ro
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 4      Routes : 4

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
 127.0.0.0/8        Direct  0    0       D  127.0.0.1   InLoopBack0
 127.0.0.1/32       Direct  0    0       D  127.0.0.1   InLoopBack0
127.255.255.255/32 Direct  0    0       D  127.0.0.1   InLoopBack0
255.255.255.255/32 Direct  0    0       D  127.0.0.1   InLoopBack0

[R1]
```

Шаг 3 Настройте IP-адреса для физических интерфейсов

Настройте IP-адреса для физических интерфейсов на основе следующей таблицы

Табл. 2-1 IP-адреса физических интерфейсов

Маршрутизатор	Интерфейс	IP-адрес/маска
R1	GigabitEthernet0/0/1	10.0.13.1/24
	GigabitEthernet0/0/0	10.0.12.1/24
R2	GigabitEthernet0/0/0	10.0.12.2/24
	GigabitEthernet0/0/2	10.0.23.2/24
R3	GigabitEthernet0/0/1	10.0.13.3/24
	GigabitEthernet0/0/2	10.0.23.3/24

R1

```
<R1>sys
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]int
[R1]interface G
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip add
[R1-GigabitEthernet0/0/1]ip address 10.0.13.1 24
Sep 17 2026 02:56:15-08:00 R1 %%%01IFNET/4/LINK_STATE(1)[0]:The line protocol IP
on the interface GigabitEthernet0/0/1 has entered the UP state.
[R1-GigabitEthernet0/0/1]quit
[R1]int
[R1]interface G
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip
[R1-GigabitEthernet0/0/0]ip add
```

```
[R1-GigabitEthernet0/0/0]ip address 10.0.12.1 24
Sep 17 2026 02:57:27-08:00 R1 %%01IFNET/4/LINK_STATE(l)[1]:The line protocol IP
on the interface GigabitEthernet0/0/0 has entered the UP state.
[R1-GigabitEthernet0/0/0]quit
[R1]
```

R2

```
<R2>sys
<R2>system-view
Enter system view, return user view with Ctrl+Z.
[R2]in
[R2]info-center
[R2]interface G
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ip add
[R2-GigabitEthernet0/0/0]ip address 10.0.12.2 24
Sep 17 2026 02:59:00-08:00 R2 %%01IFNET/4/LINK_STATE(l)[0]:The line protocol IP
on the interface GigabitEthernet0/0/0 has entered the UP state.
[R2-GigabitEthernet0/0/0]quit
[R2]in
[R2]info-center
[R2]interface G
[R2]interface GigabitEthernet 0/0/2
[R2-GigabitEthernet0/0/2]ip ad
[R2-GigabitEthernet0/0/2]ip address 10.0.23.2 24
Sep 17 2026 02:59:46-08:00 R2 %%01IFNET/4/LINK_STATE(l)[1]:The line protocol IP
on the interface GigabitEthernet0/0/2 has entered the UP state.
[R2-GigabitEthernet0/0/2]quit
[R2]
```

R3

```
<R3>sys
<R3>system-view
Enter system view, return user view with Ctrl+Z.
[R3]in
[R3]info-center
[R3]interface G
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]ip add
[R3-GigabitEthernet0/0/1]ip address 10.0.13.3 24
[R3-GigabitEthernet0/0/1]
Sep 17 2026 03:00:37-08:00 R3 %%01IFNET/4/LINK_STATE(l)[0]:The line protocol IP
on the interface GigabitEthernet0/0/1 has entered the UP state.
[R3-GigabitEthernet0/0/1]quit
[R3]int
[R3]interface G
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]ip add
[R3-GigabitEthernet0/0/2]ip address 10.0.23.3 24
Sep 17 2026 03:01:06-08:00 R3 %%01IFNET/4/LINK_STATE(l)[1]:The line protocol IP
on the interface GigabitEthernet0/0/2 has entered the UP state.
[R3-GigabitEthernet0/0/2]quit
[R3]
```

Проверьте наличие связи с помощью инструмента ping.

R1

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]ping 10.0.13.3
PING 10.0.13.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=140 ms
  Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=20 ms
  Reply from 10.0.13.3: bytes=56 Sequence=4 ttl=255 time=20 ms
  Reply from 10.0.13.3: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.13.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/46/140 ms

[R1]ping 10.0.12.2
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=110 ms
  Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=20 ms
  Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=10 ms
  Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=10 ms
  Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/34/110 ms

[R1]
```

R2

```
<R2>system-view
Enter system view, return user view with Ctrl+Z.
[R2]ping 10.0.12.1
PING 10.0.12.1: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=255 time=20 ms
  Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=255 time=20 ms
  Reply from 10.0.12.1: bytes=56 Sequence=3 ttl=255 time=20 ms
  Reply from 10.0.12.1: bytes=56 Sequence=4 ttl=255 time=20 ms
  Reply from 10.0.12.1: bytes=56 Sequence=5 ttl=255 time=10 ms

--- 10.0.12.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/18/20 ms

[R2]ping 10.0.23.3
PING 10.0.23.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=255 time=40 ms
```

```
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=255 time=30 ms
Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=255 time=30 ms
Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=255 time=20 ms
Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=255 time=10 ms
```

--- 10.0.23.3 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 10/26/40 ms

[R2]

R3

<R3>system-view

Enter system view, return user view with Ctrl+Z.

[R3]ping 10.0.13.1

PING 10.0.13.1: 56 data bytes, press CTRL_C to break

Reply from 10.0.13.1: bytes=56 Sequence=1 ttl=255 time=30 ms

Reply from 10.0.13.1: bytes=56 Sequence=2 ttl=255 time=30 ms

Reply from 10.0.13.1: bytes=56 Sequence=3 ttl=255 time=20 ms

Reply from 10.0.13.1: bytes=56 Sequence=4 ttl=255 time=20 ms

Reply from 10.0.13.1: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.13.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 20/24/30 ms

[R3]ping 10.0.23.2

PING 10.0.23.2: 56 data bytes, press CTRL_C to break

Reply from 10.0.23.2: bytes=56 Sequence=1 ttl=255 time=30 ms

Reply from 10.0.23.2: bytes=56 Sequence=2 ttl=255 time=20 ms

Reply from 10.0.23.2: bytes=56 Sequence=3 ttl=255 time=20 ms

Reply from 10.0.23.2: bytes=56 Sequence=4 ttl=255 time=20 ms

Reply from 10.0.23.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.23.2 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 20/22/30 ms

[R3]

Выведите на экран таблицу маршрутизации R1.

R1

[R1]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 10 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
------------------	-------	-----	------	-------	---------	-----------

10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet0/0/0
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/1
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Шаг 4 Создайте loopback-интерфейс.

Настройте loopback-интерфейс в соответствии со следующей таблицей.

Табл. 2-2 IP-адреса loopback-интерфейсов

Маршрутизатор	Интерфейс	IP-адрес/маска
R1	LoopBack0	10.0.1.1/32
R2	LoopBack0	10.0.1.2/32
R3	LoopBack0	10.0.1.3/32

R1

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]int
[R1]interface L
[R1]interface LoopBack0
[R1-LoopBack0]ip add
[R1-LoopBack0]ip address 10.0.1.1 32
[R1-LoopBack0]
```

R2

```
<R2>system-view
Enter system view, return user view with Ctrl+Z.
[R2]in
[R2]info-center
[R2]interface l
[R2]interface LoopBack0
[R2-LoopBack0]ip add
[R2-LoopBack0]ip address 10.0.1.2 32
[R2-LoopBack0]
```

R3

```
<R3>system-view
Enter system view, return user view with Ctrl+Z.
[R3]in
[R3]info-center
[R3]interface l
[R3]interface LoopBack0
[R3-LoopBack0]ip a
```

```
[R3-LoopBack0]ip address 10.0.1.3 32
[R3-LoopBack0]
```

Выведите на экран таблицу маршрутизации на маршрутизаторе

R1

```
[R1-LoopBack0]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 11    Routes : 11

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
 10.0.1.1/32 Direct  0    0       D  127.0.0.1   LoopBack0
 10.0.12.0/24 Direct  0    0       D  10.0.12.1   GigabitEthernet
0/0/0
 10.0.12.1/32 Direct  0    0       D  127.0.0.1   GigabitEthernet
0/0/0
 10.0.12.255/32 Direct  0    0       D  127.0.0.1   GigabitEthernet
0/0/0
 10.0.13.0/24 Direct  0    0       D  10.0.13.1   GigabitEthernet
0/0/1
 10.0.13.1/32 Direct  0    0       D  127.0.0.1   GigabitEthernet
0/0/1
 10.0.13.255/32 Direct  0    0       D  127.0.0.1   GigabitEthernet
0/0/1
 127.0.0.0/8 Direct  0    0       D  127.0.0.1   InLoopBack0
 127.0.0.1/32 Direct  0    0       D  127.0.0.1   InLoopBack0
127.255.255.255/32 Direct  0    0       D  127.0.0.1   InLoopBack0
255.255.255.255/32 Direct  0    0       D  127.0.0.1   InLoopBack0

[R1-LoopBack0]
```

R2

```
[R2-LoopBack0]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 11    Routes : 11

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
 10.0.1.2/32 Direct  0    0       D  127.0.0.1   LoopBack0
 10.0.12.0/24 Direct  0    0       D  10.0.12.2   GigabitEthernet
0/0/0
 10.0.12.2/32 Direct  0    0       D  127.0.0.1   GigabitEthernet
0/0/0
 10.0.12.255/32 Direct  0    0       D  127.0.0.1   GigabitEthernet
0/0/0
 10.0.23.0/24 Direct  0    0       D  10.0.23.2   GigabitEthernet
0/0/2
 10.0.23.2/32 Direct  0    0       D  127.0.0.1   GigabitEthernet
0/0/2
```

```
10.0.23.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/2
127.0.0.0/8 Direct 0 0 D 127.0.0.1 InLoopBack0
127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
```

[R2-LoopBack0]

R3

```
[R3-LoopBack0]display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public

Destinations : 11 Routes : 11

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.3/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.13.0/24	Direct	0	0	D	10.0.13.3	GigabitEthernet
0/0/1						
10.0.13.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
10.0.23.0/24	Direct	0	0	D	10.0.23.3	GigabitEthernet
0/0/2						
10.0.23.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R3-LoopBack0]

Проверьте наличие связи между loopback-интерфейсами

R1

```
[R1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.1.2 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

[R1]

R2

```
[R2]ping -a 10.0.1.2 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.1.1 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss

[R2]
```

R3

```
[R3]ping -a 10.0.1.3 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.1.1 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss

[R3]
```

* На данный момент у маршрутизатора нет маршрута к IP-адресу пункта назначения. Таким образом, операция ping не выполняется.

Шаг 5 Настройте статические маршруты.

На маршрутизаторе R1 настройте маршрут к интерфейсам LoopBack0 маршрутизаторов R2 и R3.

R1

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]ip ro
[R1]ip route
[R1]ip route-static 10.0.1.2 32 10.0.12.2
[R1]ip route-static 10.0.1.3 32 10.0.13.3
```

Выведите на экран таблицу маршрутизации R1.

R1

```
[R1]display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.2/32	Static	60	0	RD	10.0.12.2	GigabitEthernet
0/0/0						
10.0.1.3/32	Static	60	0	RD	10.0.13.3	GigabitEthernet
0/0/1						
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet
0/0/0						
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet
0/0/1						
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R1]

Проверьте возможность установления связи.

R1

```
[R1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.1.2 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

[R1]

* Связь с интерфейсом LoopBack0 маршрутизатора R2 по-прежнему отсутствует, поскольку у R2 нет маршрута к интерфейсу LoopBack0 маршрутизатора R1.

На R2 добавьте маршрут к интерфейсу LoopBack0 маршрутизатора R1.

R2

```
[R2]ip route-static 10.0.1.1 32 10.0.12.1
```

Проверьте возможность установления связи.

```
[R1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=255 time=10 ms
Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=255 time=30 ms
Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=255 time=30 ms
Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=255 time=30 ms
Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.1.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 10/24/30 ms

[R1]
```

LoopBack0 на R1 может взаимодействовать с LoopBack0 на R2.

Настройте другие необходимые маршруты.

R2

```
[R2]ip route-static 10.0.1.3 32 10.0.23.3
```

R3

```
[R3]ip route-static 10.0.1.1 32 10.0.13.1
[R3]ip route-static 10.0.1.2 32 10.0.23.2
```

Проверьте возможность установления связи между интерфейсами LoopBack0 маршрутизаторов

R2

```
[R2]ping -a 10.0.1.2 10.0.1.3
PING 10.0.1.3: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.3: bytes=56 Sequence=1 ttl=255 time=40 ms
Reply from 10.0.1.3: bytes=56 Sequence=2 ttl=255 time=30 ms
Reply from 10.0.1.3: bytes=56 Sequence=3 ttl=255 time=30 ms
Reply from 10.0.1.3: bytes=56 Sequence=4 ttl=255 time=10 ms
Reply from 10.0.1.3: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.1.3 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 10/26/40 ms

[R2]
```

R3

```
[R3]ping -a 10.0.1.3 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=255 time=20 ms
Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=255 time=30 ms
Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=255 time=20 ms
Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=255 time=40 ms
Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.1.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 20/28/40 ms
```

```
[R3]ping -a 10.0.1.3 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=255 time=30 ms
Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=255 time=10 ms
Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=255 time=20 ms
Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=255 time=20 ms
Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.1.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 10/20/30 ms
```

```
[R3]
```

Шаг 6 Настройте маршрут от R1 к R2 через R3 в качестве резервного маршрута от LoopBack0 R1 к LoopBack0 R2.

Настройте статические маршруты на R1 и R2.

R1

```
[R1]ip route-static 10.0.1.2 32 10.0.13.3 preference 100
```

R2

```
[R2]ip route-static 10.0.1.1 32 10.0.23.3 preference 100
```

Выведите на экран таблицы маршрутизации R1 и R2.

R1

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 13      Routes : 13

Destination/Mask    Proto   Pre  Cost   Flags NextHop        Interface
-----
10.0.1.1/32 Direct  0     0      D    127.0.0.1    LoopBack0
```

```

10.0.1.2/32 Static 60 0 RD 10.0.12.2 GigabitEthernet
0/0/0
10.0.1.3/32 Static 60 0 RD 10.0.13.3 GigabitEthernet
0/0/1
10.0.12.0/24 Direct 0 0 D 10.0.12.1 GigabitEthernet
0/0/0
10.0.12.1/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/0
10.0.12.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/0
10.0.13.0/24 Direct 0 0 D 10.0.13.1 GigabitEthernet
0/0/1
10.0.13.1/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/1
10.0.13.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet
0/0/1
127.0.0.0/8 Direct 0 0 D 127.0.0.1 InLoopBack0
127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

```

[R1]

R2

```

[R2]display ip routing-table
Route Flags: R - relay, D - download to fib

```

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Static	60	0	RD	10.0.12.1	GigabitEthernet
0/0/0						
10.0.1.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.3/32	Static	60	0	RD	10.0.23.3	GigabitEthernet
0/0/2						
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GigabitEthernet
0/0/0						
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GigabitEthernet
0/0/2						
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R2]

* Статический маршрут со значением предпочтения 100 не был добавлен в таблицу маршрутизации.

Отключите интерфейс GigabitEthernet0/0/0 на маршрутизаторах R1 и R2, чтобы сделать недействительным маршрут с наивысшим приоритетом.

R1

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]shut
[R1-GigabitEthernet0/0/0]shutdown
Sep 17 2026 04:41:45-08:00 R1 %%%01IFPDT/4/IF_STATE(l)[0]:Interface GigabitEthern
et0/0/0 has turned into DOWN state.
[R1-GigabitEthernet0/0/0]
[R1-GigabitEthernet0/0/0]
Sep 17 2026 04:41:46-08:00 R1 %%%01IFNET/4/LINK_STATE(l)[1]:The line protocol IP
on the interface GigabitEthernet0/0/0 has entered the DOWN state.
[R1-GigabitEthernet0/0/0]
```

R2

```
[R2]
Sep 17 2026 04:41:52-08:00 R2 %%%01IFPDT/4/IF_STATE(l)[0]:Interface GigabitEthern
et0/0/0 has turned into DOWN state.
[R2]
Sep 17 2026 04:41:52-08:00 R2 %%%01IFNET/4/LINK_STATE(l)[1]:The line protocol IP
on the interface GigabitEthernet0/0/0 has entered the DOWN state.
[R2]
```

Выведите на экран таблицы маршрутизации на R1 и R2. Из командного вывода видно, что маршруты с более низким приоритетом активируются, когда маршруты с более высоким приоритетом становятся недействительными.

R1

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 10    Routes : 10

Destination/Mask  Proto  Pre  Cost    Flags NextHop        Interface
10.0.1.1/32 Direct 0   0      D  127.0.0.1    LoopBack0
10.0.1.2/32 Static 100  0      RD  10.0.13.3    GigabitEthernet
0/0/1
10.0.1.3/32 Static 60   0      RD  10.0.13.3    GigabitEthernet
0/0/1
10.0.13.0/24 Direct 0   0      D  10.0.13.1    GigabitEthernet
0/0/1
10.0.13.1/32 Direct 0   0      D  127.0.0.1    GigabitEthernet
0/0/1
10.0.13.255/32 Direct 0   0      D  127.0.0.1    GigabitEthernet
0/0/1
127.0.0.0/8 Direct 0   0      D  127.0.0.1    InLoopBack0
127.0.0.1/32 Direct 0   0      D  127.0.0.1    InLoopBack0
127.255.255.255/32 Direct 0   0      D  127.0.0.1    InLoopBack0
255.255.255.255/32 Direct 0   0      D  127.0.0.1    InLoopBack0

[R1]
```

R2

```
[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 10      Routes : 10

Destination/Mask  Proto  Pre  Cost    Flags NextHop        Interface

 10.0.1.1/32 Static 100  0      RD  10.0.23.3    GigabitEthernet
0/0/2
 10.0.1.2/32 Direct 0   0      D  127.0.0.1    LoopBack0
 10.0.1.3/32 Static 60   0      RD  10.0.23.3    GigabitEthernet
0/0/2
 10.0.23.0/24 Direct 0   0      D  10.0.23.2    GigabitEthernet
0/0/2
 10.0.23.2/32 Direct 0   0      D  127.0.0.1    GigabitEthernet
0/0/2
 10.0.23.255/32 Direct 0   0      D  127.0.0.1    GigabitEthernet
0/0/2
 127.0.0.0/8 Direct 0   0      D  127.0.0.1    InLoopBack0
 127.0.0.1/32 Direct 0   0      D  127.0.0.1    InLoopBack0
127.255.255.255/32 Direct 0   0      D  127.0.0.1    InLoopBack0
255.255.255.255/32 Direct 0   0      D  127.0.0.1    InLoopBack0

[R2]
```

В этом случае исходный статический маршрут становится недействительным и активируется статический маршрут с более низким приоритетом.

Проверьте возможность установления связи.

R1

```
[R1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=254 time=50 ms
Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=254 time=30 ms
Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=254 time=30 ms
Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=254 time=20 ms
Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=254 time=40 ms

--- 10.0.1.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 20/34/50 ms

[R1]
```

R2

```
[R2]ping -a 10.0.1.2 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=254 time=20 ms
Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=254 time=40 ms
Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=254 time=20 ms
Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=254 time=30 ms
Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=254 time=30 ms
```

```
--- 10.0.1.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 20/28/40 ms
```

[R2]

Выполните трассировку маршрута, по которому передаются пакеты данных.

R1

```
[R1]tracert -a 10.0.1.1 10.0.1.2
```

```
traceroute to 10.0.1.2(10.0.1.2), max hops: 30 ,packet length: 40,press CTRL_C
to break
```

```
1 10.0.13.3 10 ms 10 ms 10 ms
```

```
2 10.0.23.2 30 ms 30 ms 20 ms
```

[R1]

R2

```
[R2]tracert -a 10.0.1.2 10.0.1.1
```

```
traceroute to 10.0.1.1(10.0.1.1), max hops: 30 ,packet length: 40,press CTRL_C
to break
```

```
1 10.0.23.3 20 ms 20 ms 20 ms
```

```
2 10.0.13.1 50 ms 30 ms 30 ms
```

[R2]

Шаг 7 Настройте маршруты по умолчанию для установления связи между интерфейсом LoopBack0 маршрутизатора R1 и интерфейсом LoopBack0 маршрутизатора R2.

Включите интерфейсы и удалите настроенные маршруты.

R1

```
[R1]interface GigabitEthernet 0/0/0
```

```
[R1-GigabitEthernet0/0/0]un
```

```
[R1-GigabitEthernet0/0/0]undo shu
```

```
[R1-GigabitEthernet0/0/0]undo shutdown
```

```
[R1-GigabitEthernet0/0/0]
```

```
Sep 17 2026 04:53:08-08:00 R1 %%01IFPDT/4/IF_STATE(1)[2]:Interface GigabitEthernet0/0/0 has turned into UP state.
```

```
[R1-GigabitEthernet0/0/0]
Sep 17 2026 04:53:08-08:00 R1 %%01IFNET/4/LINK_STATE(l)[3]:The line protocol IP
on the interface GigabitEthernet0/0/0 has entered the UP state.
[R1-GigabitEthernet0/0/0]
[R1-GigabitEthernet0/0/0]quit
[R1]undo ip route-static 10.0.1.2 32 10.0.12.2
[R1]undo ip route-static 10.0.1.2 32 10.0.13.3 preference 100
```

Выведите на экран таблицу маршрутизации R1.

R1

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 12    Routes : 12

Destination/Mask    Proto  Pre  Cost    Flags NextHop         Interface
10.0.1.1/32 Direct 0   0      D  127.0.0.1    LoopBack0
10.0.1.3/32 Static 60   0      RD  10.0.13.3    GigabitEthernet
0/0/1
10.0.12.0/24 Direct 0   0      D  10.0.12.1    GigabitEthernet
0/0/0
10.0.12.1/32 Direct 0   0      D  127.0.0.1    GigabitEthernet
0/0/0
10.0.12.255/32 Direct 0   0      D  127.0.0.1    GigabitEthernet
0/0/0
10.0.13.0/24 Direct 0   0      D  10.0.13.1    GigabitEthernet
0/0/1
10.0.13.1/32 Direct 0   0      D  127.0.0.1    GigabitEthernet
0/0/1
10.0.13.255/32 Direct 0   0      D  127.0.0.1    GigabitEthernet
0/0/1
127.0.0.0/8 Direct 0   0      D  127.0.0.1    InLoopBack0
127.0.0.1/32 Direct 0   0      D  127.0.0.1    InLoopBack0
127.255.255.255/32 Direct 0   0      D  127.0.0.1    InLoopBack0
255.255.255.255/32 Direct 0   0      D  127.0.0.1    InLoopBack0

[R1]
```

R1 не имеет маршрута к LoopBack0 (10.1.1.2/32) маршрутизатора R2.

Настройте маршрут по умолчанию на R1

R1

```
[R1]ip route-static 0.0.0.0 0 10.0.12.2
```

Выведите на экран таблицу маршрутизации R1.

R1

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
```

Routing Tables: Public

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	60	0	RD	10.0.12.2	GigabitEthernet0/0/0
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.1.3/32	Static	60	0	RD	10.0.13.3	GigabitEthernet0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.1	GigabitEthernet0/0/0
10.0.12.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.13.0/24	Direct	0	0	D	10.0.13.1	GigabitEthernet0/0/1
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R1]

Маршрут по умолчанию был активирован

Проверьте наличие связи между LoopBack0 маршрутизатора R1 и LoopBack0 маршрутизатора R2.

R1

```
[R1]ping -a 10.0.1.1 10.0.1.2
PING 10.0.1.2: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.2: bytes=56 Sequence=1 ttl=255 time=30 ms
Reply from 10.0.1.2: bytes=56 Sequence=2 ttl=255 time=20 ms
Reply from 10.0.1.2: bytes=56 Sequence=3 ttl=255 time=20 ms
Reply from 10.0.1.2: bytes=56 Sequence=4 ttl=255 time=20 ms
Reply from 10.0.1.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.1.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 20/22/30 ms
```

[R1]

R2

```
<R2>sys
Enter system view, return user view with Ctrl+Z.
[R2]ping -a 10.0.1.2 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=255 time=30 ms
```

```
Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=255 time=20 ms
Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=255 time=20 ms
Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=255 time=20 ms
Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=255 time=20 ms
```

--- 10.0.1.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 20/22/30 ms

[R2]

LoopBack0 на R1 может взаимодействовать с LoopBack0 на R2.

Конфигурация на R1

[V200R003C00]

#

sysname R1

#

snmp-agent local-engineid 800007DB0300000000000000

snmp-agent

#

clock timezone China-Standard-Time minus 08:00:00

#

portal local-server load portalpage.zip

#

drop illegal-mac alarm

#

set cpu-usage threshold 80 restore 75

#

aaa

authentication-scheme default

authorization-scheme default

accounting-scheme default

domain default

domain default_admin

local-user admin password cipher %\$%\$K8m.Nt84DZ}e#<0`8bmE3Uw}%\$%\$

local-user admin service-type http

#

firewall zone Local

priority 15

#

interface GigabitEthernet0/0/0

ip address 10.0.12.1 255.255.255.0

#

interface GigabitEthernet0/0/1

ip address 10.0.13.1 255.255.255.0

#

interface GigabitEthernet0/0/2

#

interface NULL0

#

```
interface LoopBack0
 ip address 10.0.1.1 255.255.255.255
#
ip route-static 0.0.0.0 0.0.0.0 10.0.12.2
ip route-static 10.0.1.3 255.255.255.255 10.0.13.3
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
```

Конфигурация на R2

```
[V200R003C00]
#
sysname R2
#
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load portalpage.zip
#
drop illegal-mac alarm
#
set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
firewall zone Local
 priority 15
#
interface GigabitEthernet0/0/0
 ip address 10.0.12.2 255.255.255.0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
 ip address 10.0.23.2 255.255.255.0
#
interface NULL0
#
interface LoopBack0
 ip address 10.0.1.2 255.255.255.255
#
```

```
ip route-static 10.0.1.1 255.255.255.255 10.0.12.1
ip route-static 10.0.1.1 255.255.255.255 10.0.23.3 preference 100
ip route-static 10.0.1.3 255.255.255.255 10.0.23.3
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
```

Конфигурация на R3

```
[V200R003C00]
#
sysname R3
#
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load portalpage.zip
#
drop illegal-mac alarm
#
set cpu-usage threshold 80 restore 75
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
 local-user admin service-type http
#
firewall zone Local
 priority 15
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
 ip address 10.0.13.3 255.255.255.0
#
interface GigabitEthernet0/0/2
 ip address 10.0.23.3 255.255.255.0
#
interface NULL0
#
interface LoopBack0
 ip address 10.0.1.3 255.255.255.255
#
ip route-static 10.0.1.1 255.255.255.255 10.0.13.1
ip route-static 10.0.1.2 255.255.255.255 10.0.23.2
```

```
#
user-interface con 0
 authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
```

Вывод

В ходе лабораторной работы были изучены основные принципы IPv4-адресации и маршрутизации. Были настроены IPv4-адреса на интерфейсах маршрутизаторов и созданы loopback-интерфейсы, позволяющие использовать стабильные логические адреса. Также были рассмотрены принципы формирования прямых маршрутов и настройки статической маршрутизации. С помощью команды ping была проверена связность между узлами сети.